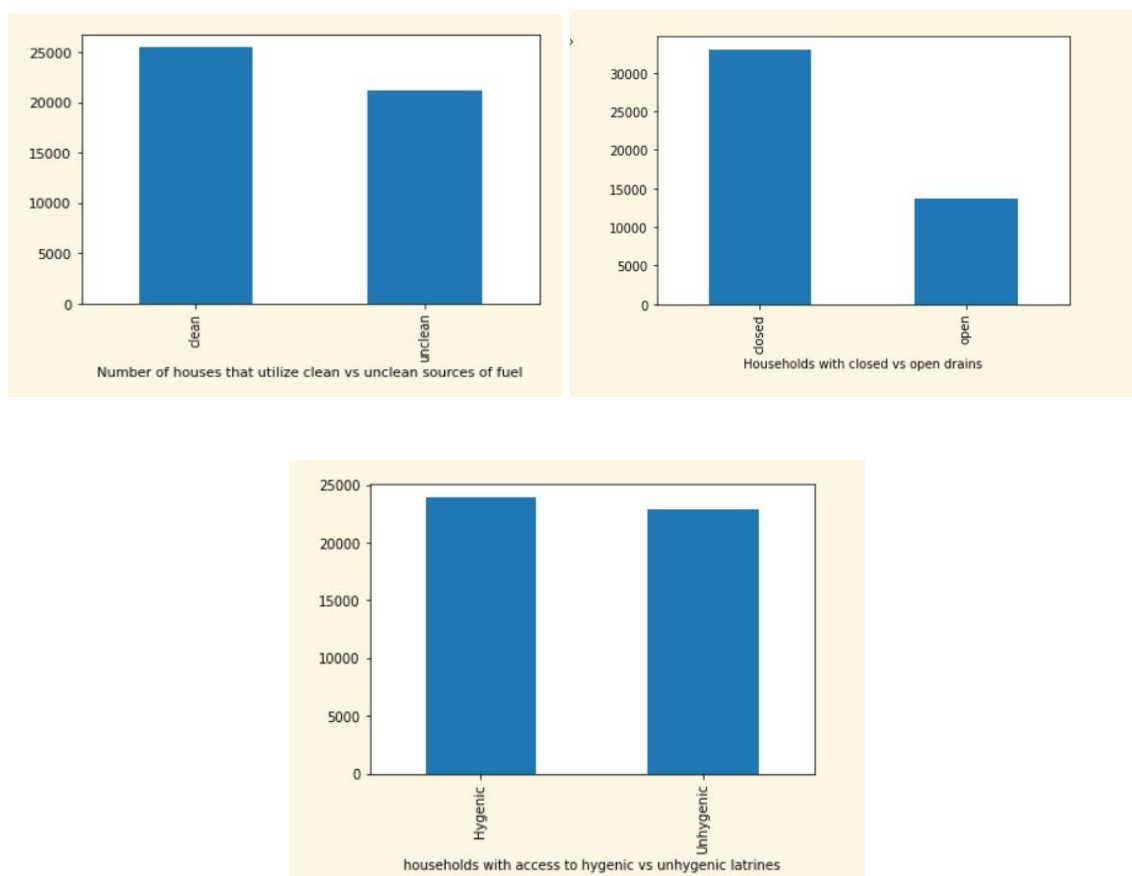


Changes in health expenditure

Neha, Vighnesh, Saranyan, Vaibhav and Vijayshree

Health expenditure outcomes of the quality of household facilities (drainage, latrine and primary source of energy for cooking) is a valid trend to be sought after when trying to understand differences between households who have access to different amenities. And yes, the underlying motive for all such analyses is that amenities are not earned but accessed, and accessibility is vital as it may have implications for the quality of life, and costs of living.

We use the social consumption data on health to understand the relationship between factors such as drainage and latrine facilities, cooking energy and health expenditure (after accounting for insurance reimbursements), while also accounting for parameters such as rural and urban as well as whether the household received treatment from a private or public hospital.



Our understanding of the trends in this All-India sample survey can be highlighted in the following observations:

- The number of households with cleaner (LPG, electricity) primary source of energy for cooking is higher

- The number of households with closed drainages is higher
- And the number of households that use septic tanks/flush latrine systems or other types that are classified as more hygienic are higher.

It should be noted that this distribution includes households in both rural and urban areas, across all states.

Moving on to the regression analysis, we try to understand if the use of unclean facilities have an effect on the health expenditure of these households. Health expenditure here accounts for reimbursements from insurance, that is deducted from the total expenses.

Multiple linear regression model

AIM: To understand the effect of household facilities on health expenditure

3 dummy variables on drainage, latrine type and primary source of energy for cooking

Dummy 1 : primary source of energy for cooking (1= traditional fuels ; 0= clean fuels(LPG and electricity))

Dummy 2 : latrine (1= service, pit, no latrine ; 0= septic tank/flush, others)

Dummy 3: drainage (1= open kutchra, open pucca, no drainage; 0= covered pucca, underground)

Dummy 4: rural =1, urban =0

Dummy 5: public =1, private = 0

Variable 6 : Monthly consumption expenditure

Model : $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \epsilon$

β_0 = the intercept

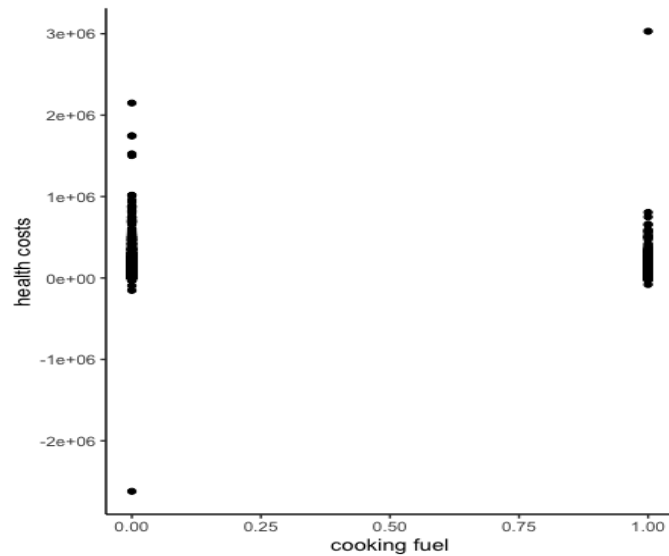
β_1 = the effect of unclean energy used during cooking

β_2 = the effect of unclean latrine

β_3 = the effect of unclean drainage

β_4 = the effect of living in a rural area

β_5 = the effect of an increase in monthly consumption expenditure.



The scatterplot above takes health expenses and places it against the use of unclean fuels- firewood and chips, charcoal, coal and coke. We notice that there is little difference between households that use both types, with outliers in both that extend to 30 lakhs worth of health expenditure. (1=unclean, 0=clean).

Model : Effect of household facilities on health expenditure

To understand the effect of the three household facilities on health expenditure, we control for rural and urban areas, whether the hospital visited was public or private and the household's monthly consumption expenditure which acts as a proxy for income levels in our analysis.

Relationship between quality of household facilities and health expenditure

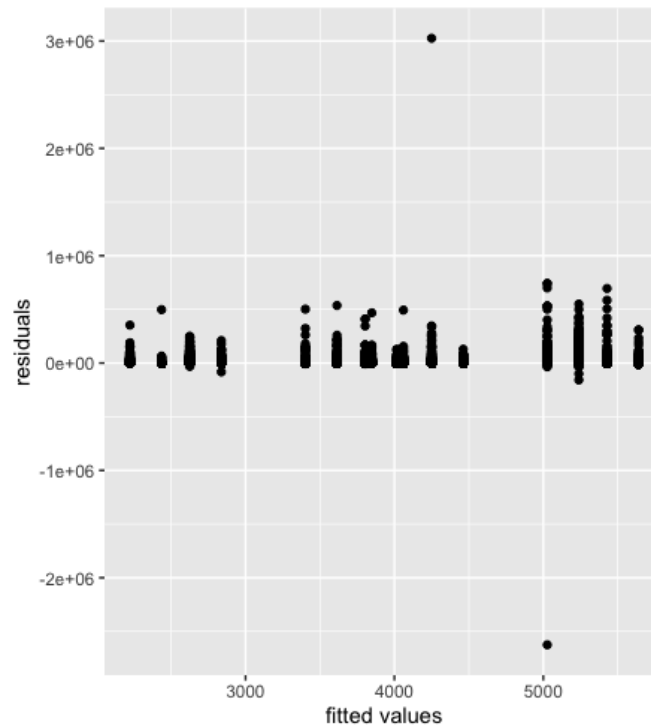
	<i>Dependent variable:</i>
	healthcosts
unclean source of energy for cooking	-1,546.246*** (454.875)
unclean latrine facilities	-1,952.787*** (449.556)
unclean drainage	1,225.646*** (468.932)
rural	679.641 (442.939)
public hospital	-15,713.280*** (387.247)
monthly CE	0.918*** (0.028)
Constant	16,341.070*** (548.868)
Observations	57,173
R ²	0.067
Adjusted R ²	0.067
Residual Std. Error	44,039.930 (df = 57166)
F Statistic	681.247*** (df = 6; 57166)
<i>Note:</i>	* p<0.1; ** p<0.05; *** p<0.01

We notice that the results for unclean latrine facilities, source of energy for cooking, type of hospital and monthly consumption expenditure are statistically significant at the 1% level. What the model tells us is that households that use unclean latrine facilities and unclean sources of energy for cooking still incur less expenses for health care compared to those that use cleaner facilities. This means that households that use traditional fuels still pay less for health care, and the positive relationship we seek between health expenditure and unclean facilities cannot be established, given there are other factors that may affect health expenditure such as the personal choice not to visit a hospital, or inability to access one, or simply cheaper options found within the public healthcare system that most of these households might approach such as PHCs or ASHA workers. Although, those that use unclean drainage systems pay around Rs.1200 more than those that don't.

We also notice that households that visit public hospitals pay Rs. 15,000 less than those that visit private hospitals.

The model also tells us that households in rural areas pay a higher amount, but this result is not statistically significant and can be set aside for this analysis.

Checking validity of this model :



There is heteroskedasticity because the residuals do not appear to cancel each other out (there are more positive residuals than there are negative) in the residuals hence failing the assumption of homoscedasticity, hence this regression wouldn't be reliable unless it is corrected for.

The Q-Q plot below shows that the residuals are normally distributed, which means that to some extent, the trends produced by the model above can be valid.

